Superstore Sales Performance Analysis

Data Storytelling & Statistical Validation

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Business Problem

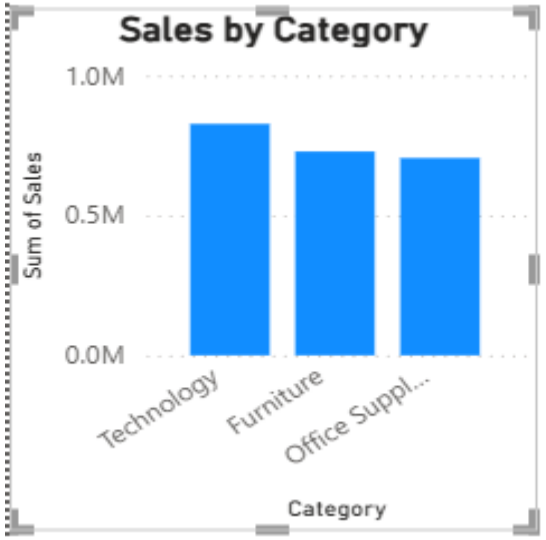
- Despite generating $2.26M in total sales with a healthy 25% profit margin, the company lacks clarity on which regions and product categories are driving profitability and which segments require strategic improvement. The business needs data-driven insights to optimize regional performance and maximize overall revenue growth.

Dataset Overview

- Total Sales: **$2.26** Million
- Total Profit: **$565.38** Thousand
- Profit Margin: **25%**
- Total Orders: **4,922**
- Average Order Value: **$459.48**
- Data covers multiple Regions, Categories, and Years (**2015–2018**)

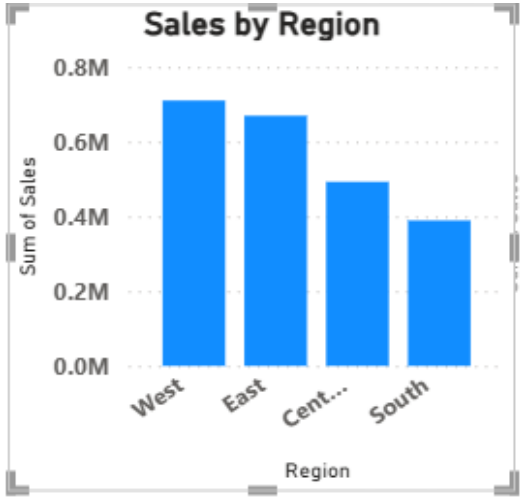
Sales Performance by Category

- Technology category generates the highest sales revenue.
- Furniture and Office Supplies contribute moderately.
- Revenue distribution shows dependency on specific product categories.
- Focusing on high-performing categories can further increase profitability.



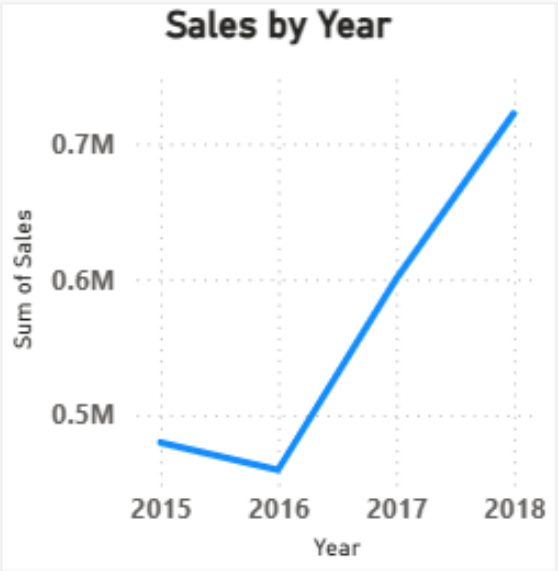
Sales Performance by Region

- The West region contributes the highest share of total sales.
- The East and Central regions show moderate performance.
- Certain regions demonstrate lower revenue contribution.
- Regional sales variation indicates potential imbalance in market performance.



Sales Trend Over Time (2015-2018)

- Sales show a consistent upward growth trend from 2015 to 2018.
- Year-on-year increase indicates expanding business performance.
- No major decline observed during the period.
- Growth trend reflects stable demand and operational efficiency.



Key Business Insights

- ✓ Revenue is primarily driven by the Technology category.
- ✓ The West region is the strongest contributor to overall sales.
- ✓ Sales performance has shown steady growth over the years.
- ✓ Profit margin of 25% indicates healthy business operations.
- ✓ However, performance differences across regions suggest potential improvement opportunities.

The observed regional variation raises an important question: Are these differences statistically significant?

Hypothesis testing – Regional sales Comparison

Objective:

- To determine whether the differences in sales across regions are statistically significant.

Null Hypothesis (H_0):

- There is no significant difference in sales across regions.

Alternative Hypothesis (H_1):

- There is a significant difference in sales across regions.

Significance Level (α): 0.05

Anova Test – regional Sales Comparison

Objective:

- To test whether there is a significant difference in sales across regions.

Test Used:

- One-Way ANOVA
- Significance Level (α) = 0.05

Key Results:

- F-value = 0.8974
- P-value = 0.4416
- F-critical = 2.6058

Decision:

- Since P-value (0.4416) > 0.05,
- we fail to reject the null hypothesis.

Conclusion:

- There is no statistically significant difference in sales among West, East, Central, and South regions.

	A	B	C	D	E	F	G	H	I	J	K	L
1	**Sales - West**	**Sales - East**	**Sales - Central**	**Sales - South**		Anova: Single Factor						
2	5.682	4.616	2.544	5.328								
3	75.792	23.36	1.788	189.588		SUMMARY						
4	28.485	10.752	3.54	6.24		*Groups*	*Count*	*Sum*	*Average*	*Variance*		
5	59.808	33.264	11.364	35.784		Sales - We	3140	710219.7	226.1846	274828.4		
6	99.696	14.352	9.264	91.2		Sales - Eas	2785	669518.7	240.4017	392334.5		
7	27.936	13.128	29.932	511.84		Sales - Cer	2277	492646.9	216.3579	404547.1		
8	18.72	106.344	121.6	46		Sales - Sou	1598	389151.5	243.5241	608166.9		
9	4.536	52.776	14.4	45.66								
10	9.156	841.568	89.52	1633.188								
11	41.72	82.344	2.724	58.34		ANOVA						
12	6.096	9.088	762.594	50.454		*Source of Varia*	*SS*	*df*	*MS*	*F*	*P-value*	*F crit*
13	99.68	8.64	19	2.88		Between G	1057278	3	352426	0.897432	0.441578	2.605816
14	12.03	125.76	91.68	15.42		Within Gro	3.85E+09	9796	392704.9			
15	21.594	25.32	30.56	113.1								
16	8.964	398.352	3.136	9.888		Total	3.85E+09	9799				
17	86.376	12.192	13.776	33.74								

Insights

- **Regional Performance**

- West, East, Central, and South regions show slight variation in average sales.
- However, ANOVA results indicate that the difference is not statistically significant.
- This means regional location alone does not strongly impact overall sales performance.

- **Sales Distribution**

- Sales are relatively balanced across all regions.
- No single region dominates overall revenue significantly.
- Performance consistency is observed across markets.

- **Variability Observation**

- Within-region sales variability is high.
- This suggests factors like product type, pricing, seasonality, or promotions may influence sales more than region.

- **Statistical Insight**

- P-value = 0.4416 (> 0.05)
- Therefore, we fail to reject the null hypothesis.
- Sales differences among regions are not statistically significant.

RECOMMENDATIONS

➢ **Uniform Strategy**

- Since there is no strong regional difference, the company can implement a standardized marketing strategy across all regions.

➢ **Focus on Other Factors**

Instead of focusing only on region, analyze:

- Product categories
- Customer segments
- Seasonal trends
- Promotional campaigns
- These may have a stronger impact on sales.

➢ **Improve Low-Performing Periods**

- Identify months or quarters with low sales and introduce targeted campaigns.

➢ **Product-Level Optimization**

- Focus on top-performing products and expand their availability across all regions.

CONCLUSION

- The One-Way ANOVA test was conducted to determine whether there is a significant difference in sales among four regions.
- The results show that:
- P-value (0.4416) is greater than the significance level (0.05).
- Therefore, we fail to reject the null hypothesis.
- This indicates that there is no statistically significant difference in sales across regions.
- Overall, regional location does not appear to be a major factor influencing sales performance. Future strategies should focus on product-level and seasonal analysis rather than geographical differences.